

Quiz — 2.1.2 Thinking Ahead — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (6 marks — 1 each) 1. **B** — binary search only works correctly on a sorted list; that is its precondition. 2. **B** — quantity and address are data *in*; A and D are outputs; C is a process. 3. **C** — storing an expensive, frequently used result for reuse is caching. 4. **A** — the cached forecast can fall out of date (stale) relative to the live source; that is the classic caching trade-off. 5. **B** — pre-tested modules speed development and carry fewer bugs. 6. **C** — dispensed cash and updated balance are results *out*; A and B are inputs, D is a process.

Section B (9 marks)

7. **[2]** Precondition: the user must be successfully authenticated/logged in (their credentials verified) before the account page is shown (1). If unchecked, an unauthenticated user could view private account data — a security/privacy breach, or the page could error on missing user data (1).
8. **[2]** Benefit (1): rates are read from fast local storage instead of re-fetching from the external service each time, so pages load faster and reduce traffic/cost. Trade-off (1): the cached rate can go stale if the market moves, so customers may be quoted an out-of-date price unless the cache is invalidated/refreshed. Must be applied to the rates scenario.
9. **[3]** Two inputs (1): item prices/quantities; any discount code or voucher entered; VAT rate. Output (1): the final total price (and any change/saving shown). Why "apply the discount" is excluded (1): it is a *process/step* that transforms data, not data flowing in or a result flowing out.
10. **[2]** Any two (1 each): faster development (write once, drop into all three apps); fewer bugs because the component is already tested; consistency — all apps validate dates the same way; easier maintenance — fix once and all apps benefit.

Section C (5 marks) 12. **[5]** One mark each, up to 5: Inputs — user ID, follow/friend list, post timestamps; Output — an ordered list of posts rendered to the feed (1). Precondition — the user must be logged in / their profile loaded before a personalised feed can be built (1). Caching — recently fetched posts or computed feed can be cached so reopening the app is fast and reduces server requests (1), accepting that the cache may go stale and need refreshing (1 — credit the trade-off if given). Reusable component — e.g. an existing date/time-formatting or image-loading or sorting module reused rather than rewritten, saving time and bugs (1). Must be applied to the feed scenario; generic definitions score poorly.

Total: 6 + 9 + 5 = 20